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Oefeningenreeks 1D nr. 9.5

$$\begin{aligned} & z^4 - 2z^2 + z + 10 \\ \Rightarrow & (3 + 2i)^4 - 2(3 + 2i)^2 + (3 + 2i) + 10 \\ = & (3 + 2i)^2 \cdot (3 + 2i)^2 - 2 \cdot (9 + 2 \cdot 3 \cdot 2i - 4) + 3 + 2i + 10 \\ = & (3 + 2i)^2 \cdot (3 + 2i)^2 - 2 \cdot (9 + 12i - 4) + 3 + 2i + 10 \\ = & (9 + 12i - 4)(9 + 12i - 4) - 10 - 24i + 3 + 2i + 10 \\ = & (5 + 12i)(5 + 12i) + 3 - 22i \\ = & 25 + 60i + 60i - 144 - 22i \\ = & -116 + 98i \end{aligned}$$

References